

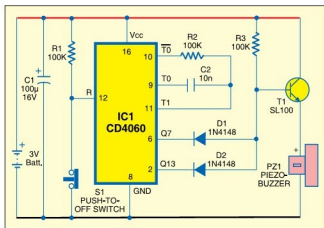
DOOR AJAR ALERT

■ T.K. HAREENDRAN

Have you ever accidentally left your front door ajar and had a pet or your little kid escape? Here is a smart solution to this prob-

lem. This circuit alerts you every ten seconds that your door is open.

The circuit is based on the popular 14-stage ripple-carry binary counter IC CD4060 and a few discrete components. The counter built around IC



CD4060 advances by one on the negative transition of each clock pulse. It is reset (which means all of its outputs go low) by applying a high pulse at its reset input pin 12 through switch S1 independent of clock. A normally-closed (N/C) push-to-off switch

S1 is used as the door sensor.

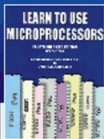
Working of the circuit is simple. When the door is closed, switch S1 opens as per the mechanical arrangement and reset pin 12 goes high. The oscillator circuit built around resistor R2 and capacitor C2 stops oscillating. Piezobuzzer PZ1 doesn't sound as outputs Q7 and Q13 remain low.

When the door is opened, switch S1 closes and reset pin 12 goes low. The oscillator starts oscillating and after a short delay of about 10 seconds, the piezobuzzer (PZ1) starts giving 10-second long beeps every 10 seconds. The beep duration of the buzzer can be varied by changing the values of timing components C2 or R2.

Assemble the circuit on a general-purpose PCB and enclose in a plastic cabinet. Mount the cabinet on the door frame and door switch S1 near the handle lock such that when the door is closed the switch (S1) opens. For powering the circuit, use an appropriate 3V battery pack. ●

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